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/* GameWin - Ludo (style Ludo Master)
   Module autonome : à charger APRÈS app.js
   <script src="ludo.js?v=1"></script>
*/
(function () {
  'use strict';
  var root =
document.getElementById('ludoGame');
  if (!root) return;

  /* ----- Données du plateau -----
  -- */
  var ORDER = ['blue', 'red', 'green',
'yellow']; // sens des aiguilles d'une
montre
  var PL = {
    blue: { label: 'Bleu', hex:
'#1e9be8', start: 39, base: [9, 0], home:
function (k) { return [14 - k, 7]; } },
    red: { label: 'Rouge', hex:
'#f0391f', start: 0, base: [0, 0], home:
function (k) { return [7, k]; } },
    green: { label: 'Vert', hex:
'#22c06a', start: 13, base: [0, 9], home:
function (k) { return [k, 7]; } },
    yellow: { label: 'Jaune', hex:
'#f6c61a', start: 26, base: [9, 9], home:
function (k) { return [7, 14 - k]; } }
  };
  var SETUPS = { 2: ['blue', 'green'], 3:

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['blue', 'red', 'green'], 4: ORDER };

var RING = [];
function add(r, c) { RING.push([r, c]); }
var i;
for (i = 1; i <= 5; i++) add(6, i);
for (i = 5; i >= 0; i--) add(i, 6);
add(0, 7); add(0, 8);
for (i = 1; i <= 5; i++) add(i, 8);
for (i = 9; i <= 14; i++) add(6, i);
add(7, 14); add(8, 14);
for (i = 13; i >= 9; i--) add(8, i);
for (i = 9; i <= 14; i++) add(i, 8);
add(14, 7); add(14, 6);
for (i = 13; i >= 9; i--) add(i, 6);
for (i = 5; i >= 0; i--) add(8, i);
add(7, 0); add(6, 0); // 52 cases

var ringAt = {}, homeAt = {}, startAt =
{};
RING.forEach(function (p, k) {
ringAt[p[0] + ',' + p[1]] = k; });
ORDER.forEach(function (c) {
startAt[PL[c].start] = c;
for (var k = 1; k <= 5; k++) { var p =
PL[c].home(k); homeAt[p[0] + ',' + p[1]] =
c; }
});
var ARROWS = { '7,0': ['→', 'red'],
'0,7': ['↓', 'green'], '7,14': ['←',
'yellow'], '14,7': ['↑', 'blue'] };

function inBase(r, c) { return (r < 6 ||
r > 8) && (c < 6 || c > 8); }

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function inCenter(r, c) { return r >= 6
&& r <= 8 && c >= 6 && c <= 8; }
function isSafe(idx) { return idx % 13
=== 0 || idx % 13 === 8; }
// rel : -1 base, 0..50 anneau, 51..55
colonne d'arrivée, 56 arrivé
function cellOf(color, rel) {
    if (rel <= 50) return
RING[(PL[color].start + rel) % 52];
    if (rel <= 55) return
PL[color].home(rel - 50);
    return null;
}

/* ----- État ----- */
var cfg = { mode: 'vs', lv: 1, n: 4,
names: ['Joueur 1', 'Joueur 2', 'Joueur 3',
'Joueur 4'] };
var S = null;          // partie en cours
var shown = false;

function later(g, fn, ms) {
setTimeout(function () { if (S === g) fn();
}, ms); }
function cur(g) { return
g.players[g.turn]; }
function nameOf(g, color) {
    for (var k = 0; k < g.players.length;
k++) if (g.players[k].color === color)
return g.players[k].name;
    return color;
}

/* ----- Style ----- */

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var css = '' +
    '.lg{max-width:540px;margin:0
auto;color:#fff;font-family:inherit}' +
    '.lg-bar{display:flex;align-
items:center;justify-content:space-
between;margin-bottom:8px;font-size:18px}' +
    +

    '.lbtn{background:#ffd21c;color:#111;border
:0;border-radius:12px;padding:12px
16px;font-weight:bold;font-
size:15px;cursor:pointer;font-
family:inherit}' +
    '.lbtn.sm{padding:8px 12px;font-
size:14px}.lbtn.alt{background:#243d62;colo
r:#fff}.lbtn.sel{background:#2bd16f;color:#
04210f}' +
    '.lg-menu{background:#0d1d36;border:1px
solid #243d62;border-
radius:18px;padding:20px;text-
align:center;display:flex;flex-
direction:column;gap:12px}' +
    '.lg-menu h2{margin:0 0
4px}.lmuted{color:#9fb3d1;margin:0}' +
    '.lmode{border:3px solid
#ffd21c;border-radius:16px;padding:20px
12px;font-size:19px;font-
weight:bold;color:#fff;cursor:pointer;font-
family:inherit}' +
    '.lmode.vs{background:linear-
gradient(135deg,#2d4fb5,#5b3fd1)}.lmode.loc
al{background:linear-
gradient(135deg,#c0262d,#e0572a)}' +
    '.lcounts{display:flex;gap:8px;justify-

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content:center}.lcounts .lbtn{flex:1}' +
    '.lrows{display:flex;flex-
direction:column;gap:8px}' +
    '.lrow{display:flex;align-
items:center;gap:10px;background:#102745;bo
rder-radius:12px;padding:8px 10px}' +
    '.ldot{width:22px;height:22px;border-
radius:50%;border:2px solid
#fff;flex:none}' +
    '.lrow input{flex:1;min-
width:0;background:#0a1730;border:1px solid
#243d62;border-
radius:8px;color:#fff;padding:9px;font-
size:15px;font-family:inherit}' +
    '.lrow input:disabled{opacity:.6}' +
    '.lg-
row{display:flex;gap:8px;margin:8px 0}' +
    '.lp{flex:1;display:flex;align-
items:center;gap:8px;padding:8px;border-
radius:14px;background:#0d1d36;border:2px
solid #243d62;min-
width:0;position:relative}' +

    '.lp.ghost{visibility:hidden}.lp.on{border-
color:var(--tc);box-shadow:0 0 14px var(--
tc)}' +
    '.lp-info{min-width:0;flex:1}.lp-
name{font-weight:bold;font-size:13px;white-
space:nowrap;overflow:hidden;text-
overflow:ellipsis}' +
    '.lp-sc{font-
size:13px;color:#ffd21c}.lp.on .lp-
name::after{content:" 👉"}' +

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'.ld{width:54px;height:54px;flex:none;border-
r-radius:12px;background:#f4f4f4;border:3px
solid #333;display:grid;grid-
template:repeat(3,1fr)/repeat(3,1fr);padding:
6px;cursor:default;opacity:.5}' +
    '.ld i{width:9px;height:9px;border-
radius:50%;place-self:center}.ld
i.on{background:#111}' +
    '.lp.on
.ld{opacity:1}.ld.rdy{cursor:pointer;animat
ion:lglow .8s infinite alternate;border-
color:#ffd21c}' +
    '@keyframes
lglow{to{transform:scale(1.12);box-shadow:0
0 14px #ffd21c}}' +
    '.lb{display:grid;grid-template-
columns:repeat(15,1fr);grid-template-
rows:repeat(15,1fr);width:100%;aspect-
ratio:1;background:#fff;border:3px solid
#20395d;border-
radius:10px;overflow:hidden;position:relati
ve}' +

'.lcell{position:relative;display:flex;flex-
-wrap:wrap;align-items:center;justify-
content:center;background:#fff;border:1px
solid #cdd6e3;box-sizing:border-box}' +

'.lcell.star::before{content:"★";position:
absolute;color:#6b7b95;font-
size:clamp(9px,3.4vw,18px);line-height:1}'
+
    '.lcell.arr::before{content:attr(data-
a);position:absolute;color:var(--ac);font-

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weight:bold;font-
size:clamp(10px,3.6vw,20px);line-height:1}'
+
    '.lbase{margin:0;display:flex;align-
items:center;justify-
content:center}.lbase.off{opacity:.35}' +
    '.lbase-
in{width:68%;height:68%;background:#fff;border-
radius:6px;display:grid;grid-template-
columns:1fr 1fr;grid-template-rows:1fr
1fr;place-items:center;padding:4%;box-
sizing:border-box}' +
    '.lslot{width:80%;height:80%;border-
radius:50%;background:rgba(0,0,0,.1);displa
y:flex;align-items:center;justify-
content:center}' +
    '.lcenter{background:conic-
gradient(from -45deg,#22c06a 0
90deg,#f6c61a 90deg 180deg,#1e9be8 180deg
270deg,#f0391f 270deg 360deg)}' +
    '.lt{width:78%;height:78%;border-
radius:50%;border:2px solid
#fff;background:radial-gradient(circle at
35% 30%,#fff8,var(--tc) 55%);box-shadow:0
2px 4px #0006;padding:0;cursor:default;box-
sizing:border-box}' +
    '.lslot
.lt{width:92%;height:92%}.lcell.multi
.lt{width:46%;height:46%}' +
    '.lt.mv{cursor:pointer;z-index:3;box-
shadow:0 0 0 3px #ffd21c;animation:lpulse
.6s infinite alternate}' +
    '@keyframes
lpulse{to{transform:scale(1.22)}}}' +

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        '.lg-msg{min-height:44px;margin:10px
0;color:#ffd21c;font-weight:bold;text-
align:center}' +
        '.lg-btns{display:flex;gap:8px;justify-
content:center;flex-wrap:wrap}' +

        '.lover{position:absolute;inset:0;background
d:#0b1630ee;display:flex;flex-
direction:column;align-
items:center;justify-
content:center;gap:10px;z-
index:5;padding:14px;text-align:center}' +
        '.lover h3{margin:0;font-
size:22px}.lover p{margin:0;font-
size:17px}';
        var st = document.createElement('style');
        st.id = 'ludo-style';
        st.textContent = css;
        document.head.appendChild(st);

        /* ----- Écrans de menu -----
        */
        function showMenu() {
            S = null;
            root.innerHTML =
                '<div class="lg"><div class="lg-
menu"><h2>🎲 Ludo</h2>' +
                '<p class="lmuted">Choisis ton mode
de jeu</p>' +
                '<button class="lmode vs" data-
act="mode" data-v="vs">🖥️ VS
Ordinateur</button>' +
                '<button class="lmode local" data-
act="mode" data-v="local">👥 Multijoueur

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local</button>' +
    '<button class="lbtn alt" data-
act="games">← Retour aux jeux</button>
</div></div>';
    }

    function showSetup() {
        S = null;
        try { if (player && player.name &&
player.name !== 'Visiteur' && cfg.names[0]
=== 'Joueur 1') cfg.names[0] = player.name;
} catch (e) {}
        var colors = SETUPS[cfg.n];
        var rows = colors.map(function (c, k) {
            var ai = cfg.mode === 'vs' && k > 0;
            var val = ai ? (k === 1 && cfg.n ===
2 ? 'Ordinateur' : 'Ordinateur ' + k) :
cfg.names[k];
            return '<div class="lrow"><span
class="ldot" style="background:' +
PL[c].hex + '"></span>' +
                '<input data-k="' + k + '"
maxlength="14" value="' + val.replace(/"/g,
'&quot;') + '" + (ai ? ' disabled' : '') +
                '></div>';
        }).join('');
        var counts = [2, 3, 4].map(function (n)
{
            return '<button class="lbtn ' + (n
=== cfg.n ? 'sel' : 'alt') + '" data-
act="count" data-v="' + n + '">' + n + '
joueurs</button>';
        }).join('');
        var lvls = [' Facile', ' Moyen',

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'● Difficile'].map(function (t, k) {
    return '<button class="lbtn ' + (k
=== cfg.lv ? 'sel' : 'alt') + '" data-
act="lvl" data-v="' + k + '">' + t +
'</button>';
}).join('');
root.innerHTML =
    '<div class="lg"><div class="lg-
menu"><h2>' + (cfg.mode === 'vs' ? '🖥 VS
Ordinateur' : '👥 Multijoueur local') +
'</h2>' +
    '<p class="lmuted">Choisis le nombre
de joueurs et les noms</p>' +
    '<div class="lcounts">' + counts +
'</div>' +
    (cfg.mode === 'vs' ? '<p
class="lmuted">Difficulté de 1\'ordinateur
(points ×1 / ×1,5 / ×2,2)</p><div
class="lcounts">' + lvls + '</div>' : '') +
    '<div class="lrows">' + rows +
'</div>' +
    '<button class="lbtn" data-act="play"
style="font-size:18px">▶ JOUER</button>' +
    '<button class="lbtn alt" data-
act="menu">← Retour</button></div></div>';
}

/* ----- Partie ----- */
function startGame() {
    var colors = SETUPS[cfg.n];
    var players = colors.map(function (c,
k) {
        var ai = cfg.mode === 'vs' && k > 0;
        return { color: c, ai: ai, name: ai ?

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(cfg.n === 2 ? 'Ordinateur' : 'Ordinateur '
+ k) : (cfg.names[k] || 'Joueur ' + (k +
1)) });
});
var g = { players: players, tok: {},
last: {}, turn: 0, dice: 0, phase: 'roll',
sixes: 0,
movable: [], rank: [], rolling:
false, msg: '' };
colors.forEach(function (c) { g.tok[c]
= [-1, -1, -1, -1]; g.last[c] = 0; });
g.msg = 'Au tour de ' + players[0].name
+ ' : lance le dé !';
S = g;
render();
if (players[0].ai) later(g, roll, 800);
}

function movableOf(g, col, v) {
var out = [], t = g.tok[col];
for (var k = 0; k < 4; k++) {
if (t[k] === -1) { if (v === 6)
out.push(k); }
else if (t[k] + v <= 56) out.push(k);
}
return out;
}

function roll() {
var g = S;
if (!g || g.phase !== 'roll' ||
g.rolling) return;
var col = cur(g).color, n = 0;
g.rolling = true;

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        var iv = setInterval(function () {
            if (S !== g) { clearInterval(iv);
return; }
            g.last[col] = 1 +
Math.floor(Math.random() * 6);
            render();
            if (++n >= 7) { clearInterval(iv);
g.rolling = false; finishRoll(g, col); }
        }, 70);
    }

    function finishRoll(g, col) {
        if (S !== g) return;
        var v = 1 + Math.floor(Math.random() *
6), p = cur(g), nm = p.name;
        g.last[col] = v; g.dice = v;
        g.sixes = v === 6 ? g.sixes + 1 : 0;
        if (g.sixes >= 3) {
            g.phase = 'wait'; g.msg = nm + ' a
fait trois 6 de suite : tour perdu.';
            render(); later(g, function () {
endTurn(g); }, 1200); return;
        }
        var mv = movableOf(g, col, v);
        g.movable = mv;
        if (!mv.length) {
            g.phase = 'wait'; g.msg = nm + ' a
fait ' + v + ' : aucun pion ne peut
bouger.';
            render(); later(g, function () {
endTurn(g); }, 1100); return;
        }
        g.phase = 'move';
        if (p.ai) { g.msg = nm + ' a fait ' + v

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+ '.'; render(); later(g, function () {
aiPlay(g); }, 800); }
    else if (mv.length === 1) { g.msg = nm
+ ' a fait ' + v + '.'; render(); later(g,
function () { doMove(g, col, mv[0]); },
500); }
    else { g.msg = nm + ' a fait ' + v + '
: choisiss un pion.'; render(); }
    }

function wouldCapture(g, col, np) {
    if (np > 50) return false;
    var idx = (PL[col].start + np) % 52;
    if (isSafe(idx)) return false;
    for (var a = 0; a < g.players.length;
a++) {
        var o = g.players[a].color; if (o ===
col) continue;
        for (var j = 0; j < 4; j++) {
            var q = g.tok[o][j];
            if (q >= 0 && q <= 50 &&
(PL[o].start + q) % 52 === idx) return
true;
        }
    }
    return false;
}

// un pion adverse peut-il atteindre
cette case de l'anneau en 1 à 6 pas ?
function threat(g, col, idx) {
    for (var a = 0; a < g.players.length;
a++) {
        var o = g.players[a].color; if (o ===

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col) continue;
    for (var j = 0; j < 4; j++) {
        var q = g.tok[o][j];
        if (q >= 0 && q <= 50) { var d =
(idx - (PL[o].start + q) % 52 + 52) % 52;
if (d >= 1 && d <= 6) return true; }
        }
    }
    return false;
}

function aiPlay(g) {
    if (S !== g || g.phase !== 'move')
return;
    var col = cur(g).color, v = g.dice,
best = -1, bs = -1e9, lv = cfg.lv;
    // Facile : joue souvent au hasard
    if (lv === 0 && Math.random() < 0.3) {
doMove(g, col,
g.movable[Math.floor(Math.random() *
g.movable.length)]); return; }
    g.movable.forEach(function (k) {
        var p = g.tok[col][k], np = p < 0 ? 0
: p + v, s = np * 0.5;
        if (np === 56) s += 100;
        if (wouldCapture(g, col, np)) s +=
90;
        if (p < 0) s += 60;
        if (np > 50 && p <= 50) s += 40;
        if (np <= 50 && isSafe((PL[col].start
+ np) % 52)) s += 25;
        if (p >= 0 && p <= 50 &&
isSafe((PL[col].start + p) % 52)) s -= 15;
        // quitter une case sûre

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        if (lv >= 1) { // Moyen et Difficile
: évite les cases où un adversaire peut
capturer, fuit le danger
            if (np <= 50 &&
!isSafe((PL[col].start + np) % 52) &&
threat(g, col, (PL[col].start + np) % 52))
s -= 45;
            if (p >= 0 && p <= 50 &&
!isSafe((PL[col].start + p) % 52) &&
threat(g, col, (PL[col].start + p) % 52)) s
+= 35;
            s += np * 0.3;
        }
        if (lv === 2 && np <= 50) { //
Difficile : poursuit les pions adverses
(menace de capture) et protège ses pions
avancés
            var ci = (PL[col].start + np) % 52;
            for (var a2 = 0; a2 <
g.players.length; a2++) {
                var oc = g.players[a2].color; if
(oc === col) continue;
                for (var j2 = 0; j2 < 4; j2++) {
                    var q2 = g.tok[oc][j2];
                    if (q2 >= 0 && q2 <= 50) { var
oi = (PL[oc].start + q2) % 52, dd = (oi -
ci + 52) % 52; if (dd >= 1 && dd <= 6 &&
!isSafe(oi)) s += 20; }
                }
            }
            if (p < 0) s += 15; // sortir vite
les pions
        }
        s += Math.random() * (lv === 2 ? 0.3

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: lv === 1 ? 1.5 : 3);
    if (s > bs) { bs = s; best = k; }
  });
  doMove(g, col, best);
}

function doMove(g, col, k) {
  if (S !== g || g.phase !== 'move' ||
g.movable.indexOf(k) < 0) return;
  var v = g.dice, t = g.tok[col], p =
t[k], np = p < 0 ? 0 : p + v, captured =
false;
  t[k] = np;
  if (np <= 50) {
    var idx = (PL[col].start + np) % 52;
    if (!isSafe(idx)) {
      g.players.forEach(function (pl) {
        if (pl.color === col) return;
        var ot = g.tok[pl.color];
        for (var j = 0; j < 4; j++) {
          if (ot[j] >= 0 && ot[j] <= 50
&& (PL[pl.color].start + ot[j]) % 52 ===
idx) { ot[j] = -1; captured = true; }
        }
      });
    }
  }
  var nm = cur(g).name, fin = np === 56,
msg = '';
  if (captured) msg = nm + ' capture un
pion !';
  if (fin) msg += nm + ' a amené un pion
à l\'arrivée !';
  if (t.every(function (x) { return x ===

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56; ))) { g.rank.push(col); msg += nm + ' a
terminé ! '; }
    g.movable = [];
    var alive = g.players.filter(function
(pl) { return g.rank.indexOf(pl.color) < 0;
});
    if (alive.length <= 1) {
        if (alive.length === 1)
g.rank.push(alive[0].color);
        g.phase = 'over'; g.msg = '';
        // Points : seulement si le joueur
humain gagne contre l'ordinateur
        var win = g.players.filter(function
(pl) { return pl.color === g.rank[0]; })
[0];
        if (cfg.mode === 'vs' && !win.ai) {
            var pts =
Math.round(g.players.length * 25 * [1, 1.5,
2.2][cfg.lv]); // × niveau
            try { player.points += pts;
saveData(); g.award = pts; } catch (e) {}
        }
        render();
        if (typeof gameOver === 'function')
gameOver(win.color === 'blue', 'Gagnant : '
+ win.name);
        return;
    }
    var extra = (v === 6 || captured ||
fin) && g.rank.indexOf(col) < 0;
    if (extra) {
        g.phase = 'roll'; g.msg = msg + nm +
' rejoue !'; render();
        if (cur(g).ai) later(g, roll, 800);

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    } else {
        g.msg = msg; g.phase = 'wait';
    render();
        later(g, function () { endTurn(g); },
msg ? 700 : 350);
    }
}

function endTurn(g) {
    if (S !== g) return;
    g.sixes = 0; g.movable = [];
    do { g.turn = (g.turn + 1) %
g.players.length; } while
(g.rank.indexOf(cur(g).color) >= 0);
    g.phase = 'roll';
    g.msg = 'Au tour de ' + cur(g).name +
(cur(g).ai ? '...' : ' : lance le dé !');
    render();
    if (cur(g).ai) later(g, roll, 800);
}

/* ----- Affichage ----- */
var PIPS = { 0: [], 1: [4], 2: [0, 8], 3:
[0, 4, 8], 4: [0, 2, 6, 8], 5: [0, 2, 4, 6,
8], 6: [0, 2, 3, 5, 6, 8] };
function dieHTML(v, rdy, c) {
    var h = '<button class="ld' + (rdy ? '
rdy' : '') + '" data-act="roll" aria-
label="Lancer le dé">';
    for (var k = 0; k < 9; k++) h += '<i' +
(PIPS[v].indexOf(k) >= 0 ? ' class="on"' :
'') + '></i>';
    return h + '</button>';
}

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function panel(g, c) {
    if (!g.tok[c]) return '<div class="lp
ghost"></div>';
    var pl = g.players.filter(function (p)
{ return p.color === c; })[0];
    var isTurn = g.phase !== 'over' &&
cur(g).color === c;
    var done = g.tok[c].filter(function (x)
{ return x === 56; }).length;
    var rdy = isTurn && g.phase === 'roll'
&& !pl.ai && !g.rolling;
    return '<div class="lp' + (isTurn ? '
on' : '') + '" style="--tc:' + PL[c].hex +
'">' +
        '<div class="lp-info"><div class="lp-
name">' + esc(pl.name) + '</div><div
class="lp-sc">🎲 ' + done + '/4</div>
</div>' +
        dieHTML(g.last[c] || 0, rdy, c) +
'</div>';
}

function esc(s) { return
String(s).replace(/[\<>"]/g, function (m) {
return { '&': '&amp;', '<': '&lt;', '>':
'&gt;', '"': '&quot;' }[m]; }); }

function tokHTML(g, c, k, mvOK) {
    var mv = mvOK && g.movable.indexOf(k)
>= 0;
    return '<button class="lt' + (mv ? '
mv' : '') + '" style="--tc:' + PL[c].hex +
'" data-c=" ' + c + ' " data-k=" ' + k + '">

```

```

</button>';
    }

    function boardHTML(g) {
        var curP = cur(g), occ = {}, h = '';
        var mvOK = g.phase === 'move' &&
!curP.ai;
        var baseTok = {};
        ORDER.forEach(function (c) {
            if (!g.tok[c]) return;
            baseTok[c] = '';
            g.tok[c].forEach(function (p, k) {
                var isCur = c === curP.color;
                if (p === -1) baseTok[c] += '<div
class="lslot">' + tokHTML(g, c, k, mvOK &&
isCur) + '</div>';
                else if (p < 56) {
                    var cell = cellOf(c, p), key =
cell[0] + ',' + cell[1];
                    (occ[key] = occ[key] ||
[]).push(tokHTML(g, c, k, mvOK && isCur));
                }
            });
        });
        ORDER.forEach(function (c) {
            var b = PL[c].base, active =
!!g.tok[c], inner = '';
            if (active) {
                // les emplacements vides (pions
sortis) restent visibles
                var slots =
(baseTok[c].match(/lslot/g) || []).length;
                inner = baseTok[c];
                for (var s = slots; s < 4; s++)

```

```

inner += '<div class="lslot"></div>';
    } else inner = '<div class="lslot">
</div><div class="lslot"></div><div
class="lslot"></div><div class="lslot">
</div>';
    h += '<div class="lbase' + (active ?
'' : ' off') + '" style="grid-area:' +
(b[0] + 1) + '/' + (b[1] + 1) + '/span
6/span 6;background:' + PL[c].hex + '">' +
    '<div class="lbase-in">' + inner +
'</div></div>';
    });
    h += '<div class="lcenter" style="grid-
area:7/7/10/10"></div>';
    for (var r = 0; r < 15; r++) for (var
c2 = 0; c2 < 15; c2++) {
        if (inBase(r, c2) || inCenter(r, c2))
continue;
        var key2 = r + ',' + c2, idx =
ringAt[key2], hc = homeAt[key2], cls =
'lcell', style = '', attr = '';
        if (hc) style = 'background:' +
PL[hc].hex + ';;';
        else if (idx !== undefined) {
            if (startAt[idx]) style =
'background:' + PL[startAt[idx]].hex + ';;';
            else if (idx % 13 === 8) cls += '
star';
        }
        if (ARROWS[key2]) { cls += ' arr';
attr = ' data-a="' + ARROWS[key2][0] + '"';
style += '--ac:' + PL[ARROWS[key2][1]].hex
+ ';;'; }
        var list = occ[key2] || [];

```

```

        if (list.length > 1) cls += ' multi';
        h += '<div class="' + cls + '"' +
        attr + ' style="grid-area:' + (r + 1) + '/' +
        + (c2 + 1) + ';" + style + '">' +
        list.join('') + '</div>';
    }
    return h;
}

function overHTML(g) {
    var medals = ['🥇', '🥈', '🥉', '4'];
    var lines = g.rank.map(function (c, k)
    { return '<p>' + medals[k] + ' ' +
    esc(nameOf(g, c)) + '</p>'; }).join('');
    return '<div class="lover"><h3>🏆
Partie terminée</h3>' + lines +
    (g.award ? '<p style="color:#ffd21c">
🎉 + ' + g.award + ' points !</p>' : '') +
    '<button class="lbtn" data-
act="again">🔄 Rejouer</button>' +
    '<button class="lbtn alt" data-
act="menu">⚙️ Changer de mode</button>
</div>';
}

function render() {
    var g = S; if (!g) return;
    root.innerHTML =
        '<div class="lg">' +
        '<div class="lg-bar"><button
class="lbtn sm alt" data-act="games">⬅️
Jeux</button><b>🎲 Ludo</b><button
class="lbtn sm alt" data-
act="menu">⚙️</button></div>' +

```

```

        '<div class="lg-row">' + panel(g,
'red') + panel(g, 'green') + '</div>' +
        '<div class="lb">' + boardHTML(g) +
(g.phase === 'over' ? overHTML(g) : '') +
'</div>' +
        '<div class="lg-row">' + panel(g,
'blue') + panel(g, 'yellow') + '</div>' +
        '<div class="lg-msg">' + esc(g.msg) +
'</div>' +
        '<div class="lg-btns"><button
class="lbtn alt" data-act="again"><img alt="refresh icon" data-bbox="608 373 628 388"/>
Nouvelle partie</button></div></div>';
    }

    /* ----- Événements ----- */
    root.addEventListener('click', function
(e) {
        var t = e.target.closest ?
e.target.closest('.lt.mv') : null;
        if (t && S) { doMove(S,
t.getAttribute('data-c'),
parseInt(t.getAttribute('data-k'), 10));
return; }
        var b = e.target.closest ?
e.target.closest('[data-act]') : null;
        if (!b) return;
        var act = b.getAttribute('data-act'), v
= b.getAttribute('data-v');
        if (act === 'roll') { if (S &&
!cur(S).ai) roll(); }
        else if (act === 'mode') { cfg.mode =
v; showSetup(); }
        else if (act === 'lvl') { cfg.lv =
parseInt(v, 10); showSetup(); }
    });

```

```

        else if (act === 'count') { cfg.n =
parseInt(v, 10); showSetup(); }
        else if (act === 'play' || act ===
'again') startGame();
        else if (act === 'menu') showMenu();
        else if (act === 'games') { S = null;
shown = false; if (typeof window.showPage
=== 'function') window.showPage('games'); }
    });
    root.addEventListener('input', function
(e) {
        var k = e.target &&
e.target.getAttribute &&
e.target.getAttribute('data-k');
        if (k !== null && k !== undefined &&
e.target.tagName === 'INPUT')
        cfg.names[parseInt(k, 10)] =
e.target.value;
    });

    /* ----- Intégration avec l'ancien
code ----- */
    function onShow() {
        if (root.classList.contains('active'))
{ if (!shown) { shown = true; showMenu(); }
}
        else { shown = false; S = null; }
    }
    new
MutationObserver(onShow).observe(root, {
attributes: true, attributeFilter:
['class'] });
    // anciennes fonctions de app.js : on les
remplace pour éviter les conflits

```

```
    window.renderLudo = function () { if
(!shown) { shown = true; showMenu(); } };
    window.startLudo = function () {
showPage('ludoGame'); };
    window.createLudoState = function () {
return {} }; };
    window.ludoChooseMode = function () {
shown = true; showMenu(); };
    window.ludoNewGame = function () { shown
= true; if (S) startGame(); else
showMenu(); };
    window.ludoRoll = function () { roll();
};
    onShow();
})();
```